
MODEL SELECTION AND VALIDATION REPORT (remove_2_biomass model)

OBSERVATIONS:1724

SANITY: OK - all models in the selection process passed sanity checks.

SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

MODEL SUMMARY (remove_2_biomass model)

Spatiotemporal model fit by REML ['sdmTMB']

Formula: fixed_formula

Time column: year

Family: lognormal(link = 'log')

Conditional model:

	coef.est	coef.se
(Intercept)	6.65	0.27
remove_2_tric	0.24	0.03
remove_2_teve	-0.36	0.02
sbt_mean	0.22	0.08
sbt_sd	-0.04	0.04
sst_sd	0.03	0.04
log_chla_mean	0.04	0.03
depth	0.25	0.04
log_sum_fishing	0.09	0.03

Dispersion parameter: 0.79

Spatiotemporal AR1 correlation (rho): 0.93

Matérn range: 342.83

Spatiotemporal marginal AR1 SD: 0.84

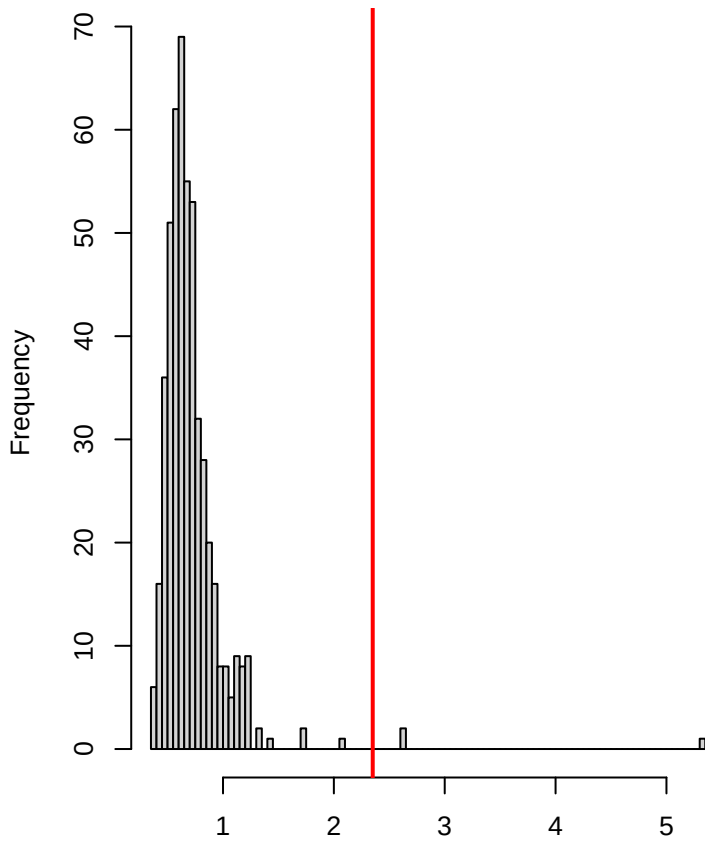
REML criterion at convergence: 12971.526

See ?tidy.sdmTMB to extract these values as a data frame.

SELECTION STATISTICS

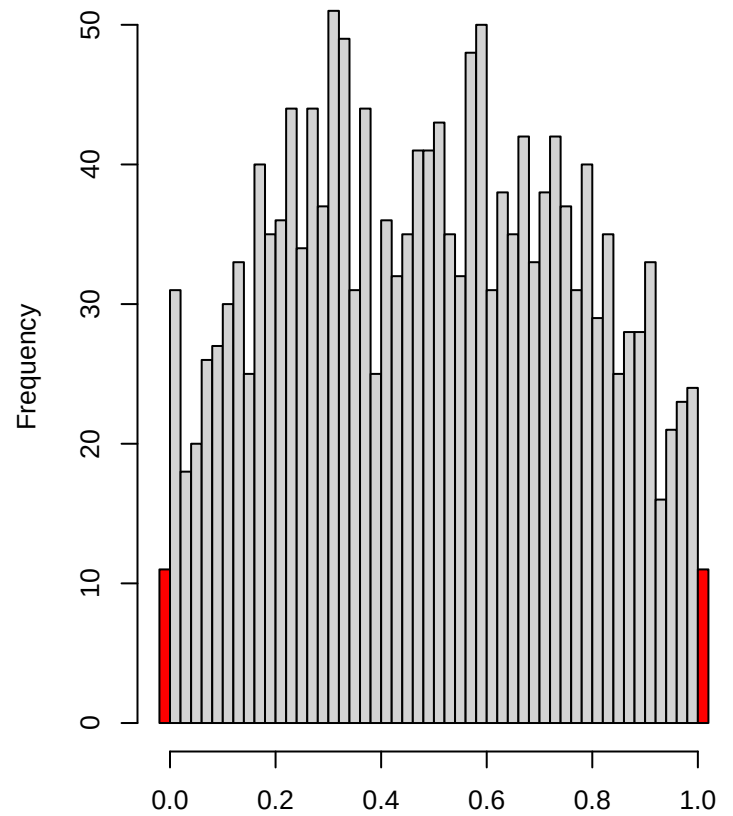
	df	AIC	delta_AIC
spatiotemporal_random_fields_ar1	13	25969.05	2.3705427
spatiotemporal_random_fields_ar1_with_anisotropy	15	25971.42	147.1155590
spatial_random_fields	12	26118.54	0.9446078
spatiotemporal_random_fields_iid_with_anisotropy	14	26119.48	0.4397381
spatiotemporal_random_fields_iid	12	26119.92	0.6750945
spatial_random_fields_with_anisotropy	14	26120.60	256.4165691
error_only	10	26377.01	NA

DHARMA nonparametric dispersion test via sd of residuals fitted vs. simulated



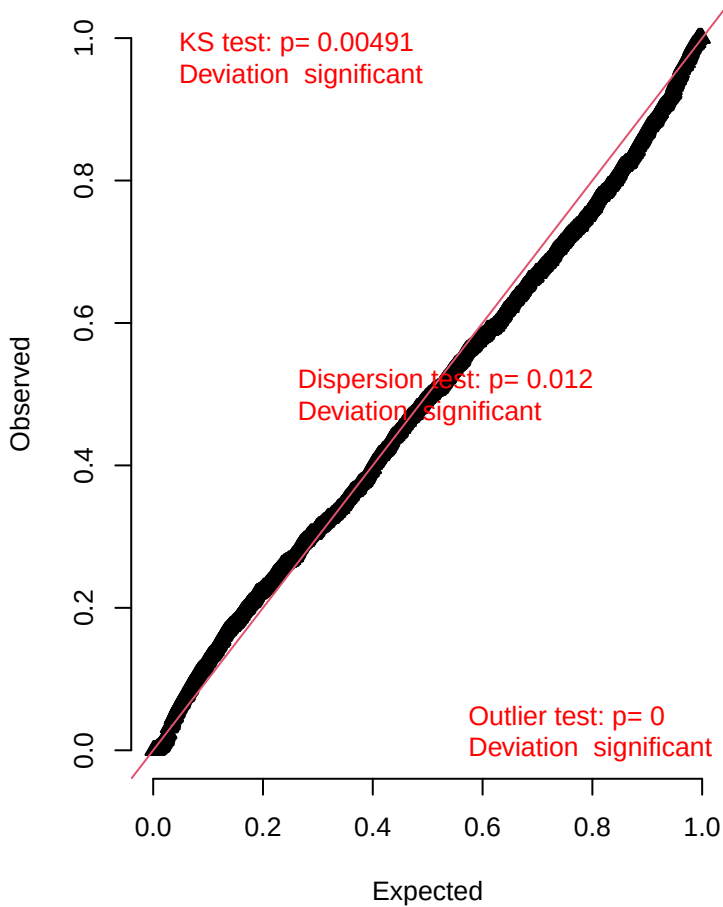
Simulated values, red line = fitted model. p-value (two.sided) =

Outlier test significant

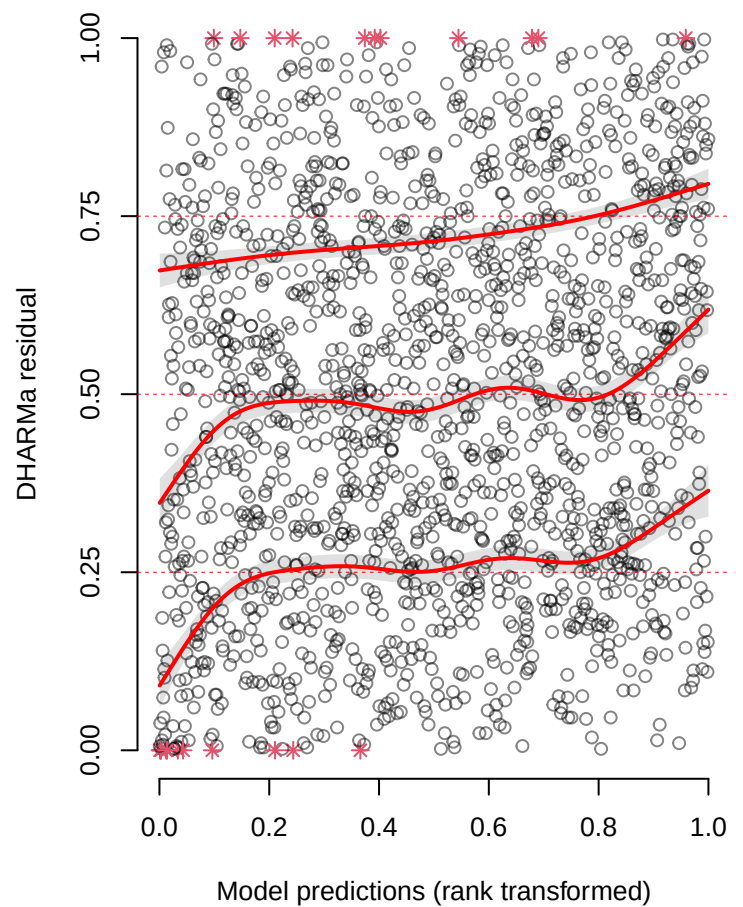


Residuals (outliers are marked red)

QQ plot residuals



DHARMA residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



DHARMa Moran's I test for distance-based autocorrelation
Moran's I = 0.028 | H1: Distance-based autocorrelation | p-value = 0.002

